

EFY Note

The source code of this project is included in this month's EFY DVD and is also available for free download at source.efymag.com



Fig. 4: Output screen of Raspi terminal window

Software

There are many operating systems designed for Raspi 2, but the most widely used are Raspian (Linux based) and Microsoft Windows 10 IoT Core. If you are a newbie to Raspi 2, check out the link: www.raspberrypi.org/help for information on Raspi. In this project, the compiler used is Python and operating system for Raspi is Raspbian. The objective is to measure the distance between the sensor and an object that is kept at an unknown distance from the sensor.

There are some packages that are required for interfacing of HC-SR04. In order to download these packages, start Terminal on Raspi and execute the commands given below. Make sure that your Raspi is hooked to the Internet, otherwise the commands will not work.

After logging on, update Raspi using the commands given below. This ensures that you have the latest software and drivers installed.

```
pi@raspberrypi ~$ sudo apt-get update
pi@raspberrypi ~$ sudo apt-get upgrade
```

Following commands get you some common Python tools:

```
pi@raspberrypi ~$ sudo apt-get install python-dev
// This should install python development tools
pi@raspberrypi ~$ sudo wget http://python.org/packages/source/R/RPi.GPIO/RPi.GPIO-0.6.1.tar.gz
// Downloads the GPIO package
pi@raspberrypi ~$ sudo tar xzf Rpi.
```

DO-IT-YOURSELF

```
GPIO-0.6.1.tar.gz
// Extracts the package
pi@raspberrypi ~$ cd Rpi.GPIO-0.6.1
//This should change the directory to Rpi GPIO folder
pi@raspberrypi ~$ sudo python setup.py install
// This will install GPIO package on Raspi
```

To make sure things are in order, reboot your Raspi by typing the following command:

```
pi@raspberrypi ~$ sudo reboot
```

Now all packages required for interfacing of HC-SR04 with Raspi have been installed.

Python code

Start Raspi and create UltraTest folder. For this tutorial, libraries used are RPi.GPIO and time. Create a new file (Ultra_Test.py) in nano editor using the commands given below:

```
pi@raspberrypi ~$ cd ~/Desktop/
UltraTest
// Change Directory to UltraTest
pi@raspberrypi ~/Desktop/UltraTest
$ sudo nano Ultra_Test.py
// Write python script as Ultra_Test.py
```

Press ctrl + o to save your Python script followed by pressing ctrl + x to exit.

Construction and Testing

After saving the code and connecting HC-SR04 to Raspi, boot Raspi. From the terminal, type the following commands to execute your program. Destination folder is UltraTest and file name is Ultra_Test.py as shown in Fig. 4.

```
pi@raspberrypi ~/Desktop/UltraTest
$ sudo python Ultra_Test.py
```

This Python script should measure the distance between the desired object and the sensor in centimetres. Refer Fig. 4 for the output screenshot of Raspi. **EFY**



Arnav Bansal is an electronics hobbyist and loves to tinker with circuit designs

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